

Math background – AI/ML Courses

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1 Probability Revision

It is recommended that you also read **Chapter 6** of *Mathematics for Machine Learning* by Marc Peter Deisenroth, A. Aldo Faisal, and Cheng Soon Ong.
<https://mml-book.github.io/>

OBJECTIVE: Revise the fundamental concepts of probability, including marginalization, conditioning, Bayes' rule and expectation.

Conditional Probability

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

where $P(A | B)$ is the **conditional probability** of A given that B occurs; $P(B)$ is the **marginal probability** of B ; and $P(A \cap B)$ is the **joint probability** of A and B . In general, from this we obtain a **chain rule**:

$$P(A_{1:n}) = P(A_n | A_{1:n-1}) \cdots P(A_2 | A_1) P(A_1)$$

Example: Assume we have an urn with 3 red balls and 1 blue ball: $U = \{r, r, r, b\}$. What is the probability of drawing (without replacement) 2 red balls in the first 2 tries?

Marginalisation

Let the sets $B_1, \dots, B_n$ be disjoint and $\bigcup_{i=1}^n B_i = \Omega$. Then

$$P(A) = \sum_{i=1}^n P(A, B_i)$$

Example: What is the probability that the second ball drawn from our urn will be red?

Bayes' Rule

Bayes' rule allows us to reverse probabilities:

$$P(A | B) = \frac{P(B | A) P(A)}{P(B)}$$

Combining this with marginalization, we obtain a powerful tool for statistical modeling:

$$P(\text{model}_i | \text{data}) = \frac{P(\text{data} | \text{model}_i) P(\text{model}_i)}{\sum_j P(\text{data} | \text{model}_j) P(\text{model}_j)}$$

That is, if we have **prior** probabilities for each model and generative data models, we can compute how likely each model is **a posteriori** (in light of our prior knowledge and the evidence brought in by the data).

Discrete Random Variables

Let E be a discrete set, e.g. $E = \{0, 1\}$. A **discrete random variable** (r.v.) is map from Ω to E :

$$X(w): \Omega \rightarrow E$$

such that for all $x \in E$, $\{w | X(w) \leq x\} \in \mathcal{F}$. Since $\mathcal{F}$ denotes the measurable sets, this condition simply means we can compute (measure) the probability $P(X = x)$.

Example: Assume we are throwing a die and are interested in the events $E = \{\text{even}, \text{odd}\}$. Here $\Omega = \{1, 2, 3, 4, 5, 6\}$. Let the r.v.

$$X(w) = \begin{cases} \text{even}, & w \in \{2, 4, 6\}, \\ \text{odd}, & w \in \{1, 3, 5\}. \end{cases}$$

We describe this r.v. with the **probability distribution** $p(x_i) = P(X = x_i) = \frac{1}{2}$, $i = 1, 2$.

The cumulative distribution function is $F(x) = P(X \leq x)$.

Example: The cumulative distribution for the die example is:

Bernoulli Random Variables

Let $E = \{0, 1\}$, with $P(X = 1) = \lambda$ and $P(X = 0) = 1 - \lambda$. Introduce the *set indicator variable*:

$$\mathbb{I}_A(w) = \begin{cases} 1, & w \in A, \\ 0, & \text{otherwise} \end{cases}$$

Using this notation, the probability distribution of a Bernoulli random variable can be written as:

$$p(x) = \lambda^{\mathbb{I}_{\{1\}}(x)} (1 - \lambda)^{\mathbb{I}_{\{0\}}(x)}.$$

Note, a particular case of the indicator function is the **Heaviside function**, named after Oliver Heaviside, also called the **step function**. $\mathbb{I}_H : \mathbb{R} \rightarrow \{0, 1\}$ where:

$$\mathbb{I}_H(x) = \begin{cases} 1, & x \geq 0, \\ 0, & x < 0 \end{cases}$$

Thus, $\mathbb{I}_H(x) = \mathbb{I}(x \geq 0)$.

Expectation of Discrete Random Variables

The expectation of a discrete random variable X is defined as

$$\mathbb{E}[X] = \sum_{x_i \in E} x_i p(x_i).$$

The expectation operator is linear, so $\mathbb{E}[ax_1 + bx_2] = a \mathbb{E}[x_1] + b \mathbb{E}[x_2]$. More generally, for any function f :

$$\mathbb{E}[f(X)] = \sum_{x_i \in E} f(x_i) p(x_i).$$

Mean $\mu = \mathbb{E}[X]$

Variance $\sigma^2 = \mathbb{E}[(X - \mu)^2]$.

For the indicator variable $\mathbb{I}_A(w)$,

$$\mathbb{E}[\mathbb{I}_A(w)] =$$

Continuous Random Variables

A random variable X is continuous if there exists a function f_X such that $f_X(x) \geq 0$ for all x , $\int_{-\infty}^{\infty} f_X(x)dx = 1$ and for every $a \leq b$,

$$P(a \leq X \leq b) = \int_a^b f_X(x)dx,$$

where the function f_X is called the **probability density function** (PDF).

Following the previous definition of **cumulative distribution function** (CDF) as $F(x) = P(X \leq x)$, we have that

$$F(x) = \int_{-\infty}^x f_X(t)dt$$

and $f_X(x) = F'(x)$ at all the points x at which F is differentiable. To simplify the notation, we will use $p(x) = f_X(x)$ as the PDF notation.

Marginals and Expectation of Continuous Random Variables

For continuous random variables, the marginal density is

$$p(x) = \int p(x, y)dy$$

Analogously, the expectation is defined as:

$$\mathbb{E}[X] = \int x p(x)dx.$$

which maintain the linear properties and can be extended for any arbitrary function f as:

$$\mathbb{E}[f(X)] = \int f(x) p(x)dx.$$

Normal (Gaussian) Random Variables

X has a Normal (Gaussian) distribution with parameters μ (mean) and σ (standard deviation), denoted by $X \sim \mathcal{N}(\mu, \sigma^2)$, if

$$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \quad x \in \mathbb{R}$$

where $\mu \in \mathbb{R}$ and $\sigma > 0$. If $\mu = 0$ and $\sigma = 1$, the variable $X \sim \mathcal{N}(0, 1)$ has a **standard Normal distribution**. The PDF and CDF of the standard Normal distribution are usually denoted as $\phi(z)$ and $\Phi(z)$ respectively.

For $Z \sim \mathcal{N}(0, 1)$, the PDF is defined as:

$$p(z) = \phi(z) =$$

Some properties of the Normal distribution:

1. If $X \sim \mathcal{N}(\mu, \sigma^2)$ and $Z = (X - \mu)/\sigma$ then $Z \sim \mathcal{N}(0, 1)$.
2. If $Z \sim \mathcal{N}(0, 1)$ and $X = \mu + \sigma Z$ then $X \sim \mathcal{N}(\mu, \sigma^2)$.
3. If $X_i \sim \mathcal{N}(\mu_i, \sigma_i^2)$ with $i = 1, \dots, n$ are independent, then

$$\sum_{i=1}^n X_i \sim \mathcal{N}\left(\sum_{i=1}^n \mu_i, \sum_{i=1}^n \sigma_i^2\right)$$

Iff (if and only if) $p(x, y)$ is a Gaussian PDF:

- $p(x)$ and $p(y)$ are Gaussian PDFs.
- $p(x|y)$ and $p(y|x)$ are Gaussian PDFs.
- Linear: $a * p(x) + b * p(y)$ is a Gaussian PDF.
- Product: $p(x) * p(y)$ is a Gaussian PDF.
- Convolution: $p(x + y)$ is a Gaussian PDF.

Multivariate Normal distribution

Thanks to the Central Limit Theorem, the multivariate normal distribution is the most used distribution in many applications. In this course we are going to use the notation: x for scalars, $\mathbf{x}$ for vectors and $\mathbf{A}$ for matrices.

A multivariate normal distribution is characterized by mean vector $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$. Then, the probability density function (PDF) is:

$$p(\mathbf{x}) = \frac{1}{(2\pi)^{d/2} |\boldsymbol{\Sigma}|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})\right)$$

More interestingly in the Machine Learning context is the log-density function that it is used for numerical reasons:

$$\log p(\mathbf{x}) = -\frac{d}{2} \log(2\pi) - \frac{1}{2} \log|\boldsymbol{\Sigma}| - \frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})$$

In this case, if we are estimating the mean vector $\boldsymbol{\mu}$, it only appears in the last term, which is a simple quadratic form (*hint: that would lead to the least-squares estimation*).

The standard multivariate Normal distribution is represented as $Z \sim \mathcal{N}(0, \mathbf{I})$. In this case:

1. If $X \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ and $Z = \boldsymbol{\Sigma}^{-1/2}(X - \boldsymbol{\mu})$ then $Z \sim \mathcal{N}(0, \mathbf{I})$.
2. If $Z \sim \mathcal{N}(0, \mathbf{I})$ and $X = \boldsymbol{\mu} + \boldsymbol{\Sigma}^{1/2}Z$ then $X \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$.
3. Because $\boldsymbol{\Sigma}$ is a symmetric and positive definite matrix, there always exist a square root matrix $\boldsymbol{\Sigma}^{1/2}$, such that:
 - $\boldsymbol{\Sigma}^{1/2}$ is also symmetric
 - $\boldsymbol{\Sigma} = \boldsymbol{\Sigma}^{1/2} \boldsymbol{\Sigma}^{1/2}$
 - $\boldsymbol{\Sigma}^{-1/2} \boldsymbol{\Sigma}^{1/2} = \boldsymbol{\Sigma}^{1/2} \boldsymbol{\Sigma}^{-1/2} = \mathbf{I}$

There are many algorithms to obtain $\Sigma^{1/2}$ like the *Cholesky decomposition*.

Suppose we partition a random Normal vector $X \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma)$ as $X = (X_a, X_b)$. We can similarly partition $\boldsymbol{\mu} = (\boldsymbol{\mu}_a, \boldsymbol{\mu}_b)$ and

$$\Sigma = \begin{pmatrix} \Sigma_{aa} & \Sigma_{ab} \\ \Sigma_{ba} & \Sigma_{bb} \end{pmatrix}$$

Then:

1. The marginal distribuion of X_a is $X_a \sim \mathcal{N}(\boldsymbol{\mu}_a, \Sigma_{aa})$.

2. The condition distribution of X_b given $X_a = x_a$ is

$$X_b|X_a = x_a \sim \mathcal{N}(\boldsymbol{\mu}_b + \Sigma_{ba}\Sigma_{aa}^{-1}(x_a - \boldsymbol{\mu}_a), \Sigma_{bb} + \Sigma_{ba}\Sigma_{aa}^{-1}\Sigma_{ab})$$

3. For matrices $\mathbf{A}$ and $\mathbf{B}$ then $\mathbf{A}X_a + \mathbf{B}X_b \sim \mathcal{N}(\mathbf{A}\boldsymbol{\mu}_a + \mathbf{B}\boldsymbol{\mu}_b, \mathbf{A}\Sigma_{aa}\mathbf{A}^\top + \mathbf{B}\Sigma_{bb}\mathbf{B}^\top)$

4. $V = (\mathbf{x} - \boldsymbol{\mu})^\top \Sigma^{-1}(\mathbf{x} - \boldsymbol{\mu}) \sim \chi_k^2$ where χ_k^2 is the *chi-squared distribution* with k degrees of freedom.

2 Information theory

OBJECTIVE: Revise the fundamental concepts of information theory that can be applied for machine learning, such as entropy and KL-divergence.

Information theory provides the mathematical framework for quantifying uncertainty, information, and the efficiency of communication. Machine learning can be seen as a form of compression as we build models to learn patterns and predictions about probability of occurrence. Similarly, a good compression algorithm should encode most likely patterns with shorter codes than unlikely appearances. For example, a language-model that is able to predict the next word is an excellent way to compress large volumes of text data.

Concepts like **entropy**, **mutual information**, and **KL-divergence** help us measure how much information is contained in data, how surprising an event is, and how different two probability distributions are. In machine learning, these tools are fundamental: they guide model training (e.g., cross-entropy loss), enable efficient representations of data, and underpin algorithms in clustering, feature selection, and generative modeling.

Entropy

In his seminal work, *A Mathematical Theory of Communication*, Claude Shannon introduce the log function as a measure of information of random variables as it have some interesting properties (most parameters in information systems scale logarithmically, it is monotonic, numerically stable, etc.). Intuitively, two identical memories in a computer should be able to encode **twice** the information of a single memory. If we use $\log_2$ the information is computed in **bits**, while in machine learning for convenience we use $\ln = \log_e$ which measure the information in **nits**. Then, we can clearly see that the *negative expected information* should be a measure of uncertainty, or lack of information.

$$H(X) \triangleq - \sum_{k=1}^K p(X = k) \log p(X = k)$$

If the variable X is continuous, each sum in this section is replaced by integral for PDFs.

Another interpretation is that the term $-\log(p(X)) = \log\left(\frac{1}{p(x)}\right)$ can be interpreted as the *surprise* of an event. An unlikely event $p(X = k) \rightarrow 0$ is unexpected when it happens, resulting in a large surprise $\log\left(\frac{1}{p(x)}\right) \rightarrow \infty$. Meanwhile, a likely event $p(X = k) \rightarrow 1$ results in no surprise when it happens $\log\left(\frac{1}{p(x)}\right) \rightarrow 0$. In this context, the **entropy is the expected surprise** which intuitively is related with the associated uncertainty (the more uncertainty, the larger surprise...).

Example: For a coin $X = (heads, tails)$ with the probability of landing heads defined as θ , that is,

$$p(X = heads) = \theta, \quad p(X = tails) = 1 - \theta$$

Then the entropy can be simplified to the **binary entropy function**,

$$H(X) = -[\theta \log p(X = heads) + (1 - \theta) \log p(X = tails)]$$

- If $\theta = 0.5$ (fair coin), $H(X) =$
- If $\theta = 1$ or $\theta = 0$ (unfair coin), $H(X) =$
- Which one is easier to predict? A fair or unfair coin?

KL-divergence

The **Kullback-Leibler divergence**, or **relative entropy** is a way to measure the dissimilarity of two probability distributions p and q . This is defined as:

$$KL(p||q) \triangleq \sum_{k=1}^K p_k \log \frac{p_k}{q_k}$$

where, again, the sum is replaced by an integral for PDFs. We can rewrite this as:

$$KL(p||q) = \sum_{k=1}^K p_k \log p_k - \sum_{k=1}^K p_k \log q_k = -H(p) + H(p, q)$$

where $H(p, q)$ is called the **cross entropy**:

$$H(p, q) = \sum_{k=1}^K p_k \log q_k$$

which is used extensively in machine learning, as it represents the average number of bits/nits¹ needed to encode data coming from a source with distribution p when we use model q .

¹The use of bits or nits depends on the log used. In classical computer science, $\log_2$ is used resulting in bits. In machine learning it is more common to use the natural logarithm $\ln$ resulting in nits.

For $p = q$, then $KL(p||q) = 0$. Thus, the KL-divergence can be used as a quantity to be minimized for getting a model q close to the data generative process d . Note that KL is called a divergence and not a distance metric, because it does not satisfy all the properties required to be a distance metric, such as being symmetric. In general, $KL(p||q) \neq KL(q||p)$.

Example: Let us have two different coins $X = (heads, tails)$ with the probability of landing heads defined as θ , like in the example before:

$$p(X = heads) = \theta, \quad p(X = tails) = 1 - \theta$$

Compute the KL-divergence of the distribution P where $\theta_P = 0.5$ (fair coin) and Q with $\theta_Q = 0.9$ (unfair coin).

$$KL(P||Q) =$$

Then, compute the reverse case (remember that KL-diverge is NOT symmetric):

$$KL(Q||P) =$$

Information gain

In many areas of machine learning (such as decision trees, active learning, etc.), it is important to keep track on the *amount of information* that we can obtain from a set of observations. For that, we can formulate the concept of **information gain**, for a random variable X , given an observation of random variable Y which took the value y :

$$IG = KL(p(X|y)||p(X)) = H(X) - H(X|y)$$

If we consider the entropy to represent the *amount of uncertainty*, the information gain represents the *amount of uncertainty* that we have removed from the system after introducing the new data point y .

Maximum-likelihood and cross-entropy

Assume the generative distribution for the data $p(x|w^*)$, where w^* represents the true parameters of the generative process. Then, the *maximum-likelihood estimator* (MLE) $\hat{w}$ minimizes the KL-divergence between $p(x|\hat{w})$ and $p(x|w^*)$.

$$\begin{aligned}
\hat{w} &= \arg \max_w \prod_{i=1}^N p(x_i|w) \\
&= \arg \max_w \sum_{i=1}^N \log p(x_i|w) && \text{*take logs, max is the same} \\
&= \arg \max_w \frac{1}{N} \sum_{i=1}^N \log p(x_i|w) - \frac{1}{N} \sum_{i=1}^N \log p(x_i|w^*) && \text{*add term } \perp w \\
&= \arg \max_w \frac{1}{N} \sum_{i=1}^N \log \frac{p(x_i|w)}{p(x_i|w^*)} && \text{*rearrange} \\
&= \arg \min_w \frac{1}{N} \sum_{i=1}^N \log \frac{p(x_i|w^*)}{p(x_i|w)} && \text{*} \max_z [f(z)] = \min_z [-f(z)] \\
&\xrightarrow{N \rightarrow \infty} \arg \min_w \int p(x|w^*) \log \frac{p(x|w^*)}{p(x|w)} dx && \text{*law of large numbers, see Monte Carlo slides} \\
&= \arg \min_w \text{KL}(p(x|w^*) || p(x|w))
\end{aligned}$$

And remember that

$$\text{KL}(p(x|w^*) || p(x|w)) = \underbrace{-H(p(x|w^*))}_{\text{entropy}} + \underbrace{H(p(x|w^*), p(x|w))}_{\text{cross-entropy}}$$

where the **entropy** is about the generative distribution, meaning that it captures the entropy on the data about the true generative process and it does not depend on the model w . Thus, the **cross-entropy** term is the only one that depends on w . Therefore, *minimizing the cross-entropy* is equivalent to maximizing the likelihood function.

Properties of MLE

The *maximum-likelihood estimator* (MLE) $\hat{w}$ is asymptotically normal. That is, as $N \rightarrow \infty$, we have:

$$\hat{w} - w^* \Rightarrow \mathcal{N}(0, F^{-1})$$

where F is the Fisher Information matrix. As a result of this, as $N \rightarrow \infty$, the expected value of the error $\mathbb{E}(\hat{w} - w^*) = 0$, meaning that the estimator is **unbiased**.

3 Optimization

It is recommended that you also read **Chapter 7** of *Mathematics for Machine Learning* by Marc Peter Deisenroth, A. Aldo Faisal, and Cheng Soon Ong.
<https://mml-book.github.io/>

OBJECTIVE: Revise some fundamental concepts of optimization theory that are used in many machine learning algorithms.

Lagrange optimization

In machine learning, we often minimize a loss function $f(w)$ (e.g., Mean Squared Error) while constraining model complexity to prevent overfitting:

$$\min_w f(w) \quad \text{subject to} \quad \|w\|_2^2 \leq C$$

where w represents the weights and $C > 0$ is the maximum allowed budget for the weight magnitude.

Using Lagrange multipliers, we convert this constrained problem into an unconstrained objective by introducing a multiplier $\lambda \geq 0$:

$$\mathcal{L}(w, \lambda) = f(w) + \lambda (\|w\|_2^2 - C)$$

When solving for w , the constant term λC does not affect the argmin. The optimization problem simplifies to:

$$\hat{w} = \arg \min_w (f(w) + \lambda \|w\|_2^2)$$

This is the standard formula for L_2 **Regularization (Ridge Regression)**, where:

- $f(w)$ forces the model to fit the training data.
- $\lambda \|w\|_2^2$ penalizes large weight magnitudes.
- λ acts as the hyperparameter controlling regularization strength.

As can be seen in Figure 1, at the optimal weight vector w^* , the gradient of the loss function and the gradient of the constraint are aligned in opposite directions:

$$\nabla f(w^*) = -\lambda \nabla (\|w^*\|_2^2) \implies \nabla f(w^*) + 2\lambda w^* = 0$$

This balance prevents the weights from growing further into the constrained region.

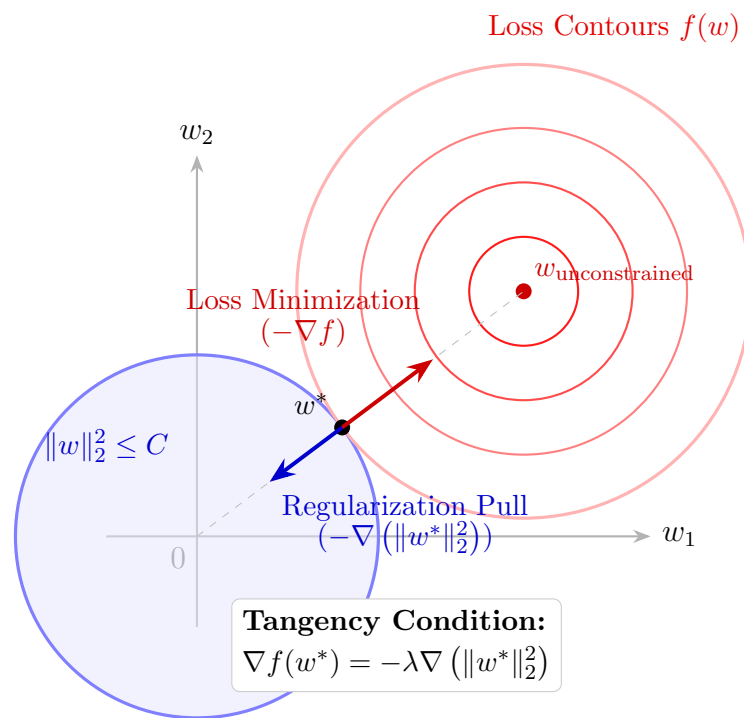


Figure 1: Gradient equilibrium of the unconstrained optimization solution.